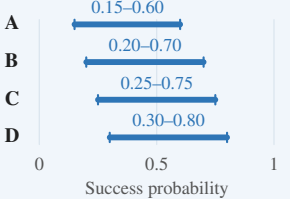
Q-BRIDGE

One shared view. Two policy questions.

Four jackets A–D · High rating = success · Reference: 25% each

Observational logs

Bounds on each success rate



Randomized experiments

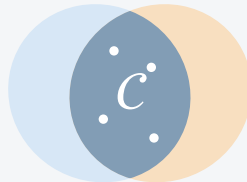
Transferred group averages

$$(m_A + m_C)/2 = 0.65$$

$$(m_B + m_D)/2 = 0.70$$

(a) Joint confidence view

All compatible success rates



Individual + group constraints hold together

Fixed data + assumptions

(b) Can this policy guarantee a gain?

Requested gain: 0.100 pp

Checked lower bound

Q1	Solve + check	+1.000 pp
Q2	Reuse + check	+0.638 pp
Q3	New solve + check	+0.500 pp

Q3: saved proof inconclusive → solve

(c) What is the best possible guarantee?

Disjoint groups: cap means → allocate → recover policy

Always C

Policy lower bound **2.500 pp** = Family upper bound **2.500 pp**

Optimal robust guarantee

Constructed example · pp = percentage points

One allowed vector: (0.60, 0.70, 0.70, 0.70)